

10th International Conference of Information and Communication Technology (ICICT-2020)

Semantic analysis based on fusion of audio/visual features for soccer video

Zengkai Wang*

College of Mathematics, Physics and Information Engineering, Jiaxing University, Jiaxing 314001, China

Abstract

In order to analysis the semantic content of soccer video, the audio/visual features are effectively extracted. Based on the theory that the variation of video content would cause the fluctuation of viewers' affection, the highlight time curve (HTC) is generated by fusion of affection arousal factors to reveal the excitement of the game. The semantic boundaries of highlights are determined by HTC combined with the domain knowledge of soccer video. With the help of distinguishable highlight feature vectors (HFVs), highlights are classified into goal, shoot, and foul. Compared with the existing works, the main contributions of this paper are as follows. We proposed a novel Hough transform based whistle detection algorithm and achieves more effective performance. A robust goalmouth detection algorithm is presented and contributed to the highlight classification phase. The highlights with semantic boundaries are accurately extracted and classified. Experiments conducted on real world soccer videos demonstrated the good performance of the proposed framework.

© 2021 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the 10th International Conference of Information and Communication Technology.

Keywords: Semantic analysis, event detection, soccer video;

1. Introduction

Video semantic analysis has been a challenging and hot-spot research area. As the wide viewer-ship and tremendous commercial potentials, soccer video has attracted number of researchers come from academia and industry. How to detect and label the highlights is the key technology of soccer video retrieval, summarization, personalized recommendation, and adaptive content transmission [1].

* Corresponding E-mail address: wangzengkai@zjxu.edu.cn

Existing works on highlight detection in soccer video could be classified into machine learning model-based approach [2], statistical-rule based approach [3], and external text-based approach [4]. Little works have been done on semantic boundaries of highlights, most existing works only detect highlight as a video shot or a fixed time-length clips, only if which contain the excitement time point [5]. Many statistical rules and machine learning model-based highlight detection methods could not evaluate the excitement degree of the highlights. The accuracy of highlights classification should also be improved. The external text-based approach faces the problems of how to precisely extract the time stamp from the video, and how to keep synchronization between texts and video events.

In this work, we present a novel soccer video semantic analysis approach based on fusion of effective audio/visual features. The framework of the proposed approach is shown in Figure 1.

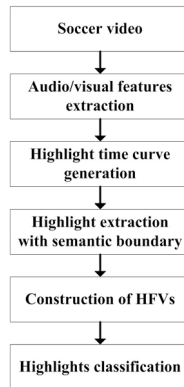


Fig. 1. Framework of the proposed approach.

The effective audio/visual features are first extracted from raw soccer video. The affection related factors are well fused to generate HTC, by which the highlights with semantic boundaries are extracted. The distinguishable HFVs are extracted from highlight segments, and fed to the trained classifier to label the highlights as goal, shoot and foul. Compared with the existing works, the main contributions of our framework are as follows.

- A novel Hough transform based whistle detection algorithm is first proposed and achieves more effective performance.
- A robust goalmouth detection algorithm is presented and contributed to the highlight classification phase.
- The semantic boundaries of highlights are determined by HTC, and the event type of highlights are accurately labeled as goal, shoot and foul.

2. Audio/visual cues detection

2.1. Extraction of audio cues

Short time energy of sound: In order to keep synchronization between audio frame and video frame, we extract 1 second audio data from the soccer video every time, and then split them into f_s frames equally, f_s is the video frame frequency per second. The short time energy of frame i , $STE(i)$ is calculated by sum up all the spectrum values of N audio samples:

$$STE(i) = \sum_{k=1}^N x^2(k) \quad (1)$$

Here, $x(k)$ is the k -th sample value in audio frames.

Whistle : Whistle in soccer game usually means foul or set-piece kick event happening. Ref. [6] utilized the zero crossing rates (ZCR) and support vector machine (SVM) classifier to detect whistle. Ref.[7] extracted Mel-Frequency Cepstral Coefficients (MFCC) and Energy features of audio, and then feed these features to a typical left-right hidden Markov Model (HMM) with five hidden states to detect whistle. These whistle detection methods rely on machine learning models which should be trained on large amount of sample data. By observing the spectrum diagram, it is found that the whistle has obvious characteristics on the spectrum diagram. Figure 2 shows the spectrum of whistle sounds in soccer video. It is easy to find that the whistle frequency usually between 3500Hz and 4500Hz, and whistle duration last about 0.1 second. So, we use Hough transform (HT) method in image processing field to detect whistle. This is different from the existing methods. The algorithm is shown in Figure 3.

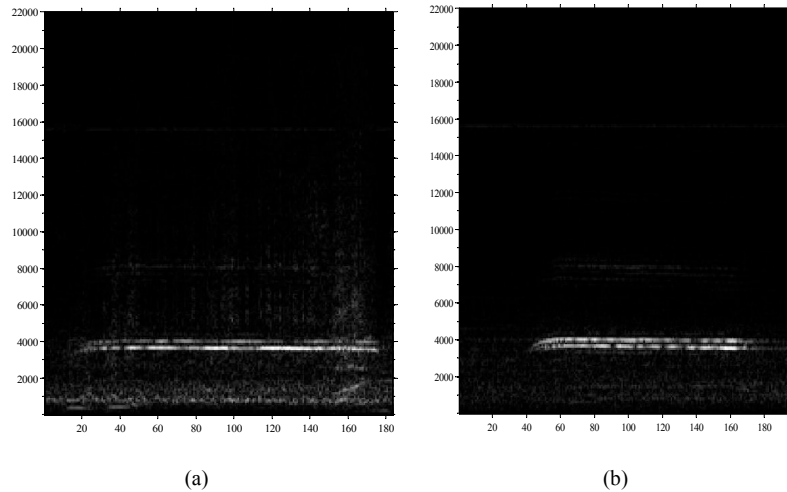


Fig. 2. Two examples of grayscale time-frequency image of whistle audio clips in soccer video. The vertical and horizontal axes denote the frequency and the number of audio frames of whistle audio clip.

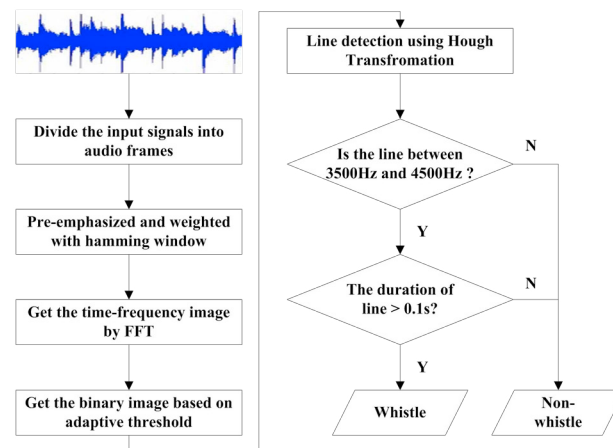


Fig. 3. Diagram of HT based whistle detection algorithm.

2.2. Extraction of visual cues

Shot type: The self-adaptive dual thresholds and dominant color percentage based shot boundary detection algorithm [8] is employed to segment the soccer video into shots. The cut and dissolve shot boundaries are

discriminated. Then, the sub-window based shot classification algorithm [9] is utilized to classify the video shots into far view shot (FVS), medium view shot (MVS), close-up shot (CUS) and out-of field shot (OFS).

Replay: Replay shot provides another opportunity to review the exciting moments, such as goals or shoots. If there is a replay shot, there is more likely a highlight event. So we choose replay shot as the feature to detect highlights. A replay is always sandwiched by a pair of logos, which indicate the start and the end of a replay. The dissolve shots are thought as the candidates of logo sequences. The reliable logo detector [10] is used to confirm whether the dissolve shots are logo sequences, which are then corrected and paired to find the replay segments.

Shot intensity: Shot intensity measures the excitement level of a shot. In soccer video, when the game is becoming intense, the magnitudes of motion activity of video frames are becoming drastic. As a primary measurement, motion activity could be employed to evaluate the game intensity of soccer video. The motion activity, $MA(i)$ at video frame i , is computed as the average magnitudes of all motion vectors:

$$MA(i) = \frac{\sum_{k=1}^B \bar{v}_k(i)}{B|\bar{v}_{\max}(i)|} \quad (2)$$

$$|\bar{v}_k(i)| = \sqrt{dx_k^2(i) + dy_k^2(i)} \quad (3)$$

Here, B is the number of blocks within a frame, $\bar{v}_k(i)$ is the motion vector of block k in frame i . $dx_k(i)$ and $dy_k(i)$ are the offsets in horizontal and vertical axes of block k in frame i , respectively. However, the amount of computation of motion activity is very large. The motion activity of a ninety minutes soccer video, whose frame rate is 25fps, is computed on the platform with Intel Xeon 2.4GHz CPU and 32GB memory, the overall computation time is about 8 hours.

By large amount of observations, we found that the shot types always related to the game intensity. In general, the far view shot, which is used to depict the whole attack/defense situations, reveals the lowest game intensity. The medium view shot, which usually means the start of attack or splendid coordination, has slightly higher game intensity. The close-up view shot and out-of field shot, which always mean some highlights have happened, have the highest game intensity. In order to reduce the computation amount, we collect large number of shots, the average motion activity of every shot is calculated and the statistical analysis method is applied for each type of shot, to get the statistical representative magnitude value of motion activity of these four types of shots respectively, i.e. shot intensity, which is expressed as:

$$SI(i) = \begin{cases} 0.54, & \text{if } shotttype(f_i) = FVS \\ 0.70, & \text{if } shotttype(f_i) = MVS \\ 0.81, & \text{if } shotttype(f_i) = CUS \\ 0.84, & \text{if } shotttype(f_i) = OFS \end{cases} \quad (4)$$

Here, $shotttype(f_i)$ denotes the shot type of video frame i , $SI(i)$ denotes shot intensity value of f_i . Readers could find more details about the calculation of shot intensity in reference [11].

Goalmouth: Goalmouth is an important cue in shoot or goal events. In soccer video, the angles and distances of camera vary frequently, the goalmouth in video frame also has different appearance. We first extract the feature vector according the characteristics of goalmouth using Algorithm 1, and then train a SVM classifier to detect the goalmouth.

Algorithm 1 Extraction of feature vector of goalmouth**Input:** Original soccer video frame**Output:** 7-dimensional feature vectors of goalmouth

- 1: Load the original soccer video frame f_0 , convert f_0 into grayscale image f_g .
- 2: Top-hat transform is carried out for f_g , which is expressed as:

$$HAT(I) = I - (I \circ s) \quad (5)$$

where I is the input grayscale image, $HAT(I)$ denotes the results top-hat transform, “ $\circ$ ” denotes morphological opening operation, s is the structure element. Denote the image of top-hat transform as f_i .

- 3: Get the binary image f_b of f_i , the threshold is set to 200.

- 4: Calculate the maximum connected length $L_i (i=1, 2, \dots, w)$ along the vertical direction for each column of f_b , where w is the width of f_b , select the two longest lines $P1, P2$ as the candidate goal posts, which satisfy

$$|P1_x - P2_x| > 18 \quad (6)$$

where $P1_x$ and $P2_x$ are horizontal coordinates of $P1$ and $P2$, respectively.

- 5: Execute connectivity analysis for f_b , continuously merge the adjacent vertical white line whose length is greater than $3L_{p1}/4$ into $P1$, and continuously merge the adjacent vertical white line whose length is greater than $3L_{p2}/4$ into $P2$, where L_{p1} and L_{p2} denote the length of candidate goal post $P1$ and $P2$, respectively.

- 6: Extract the candidate goal post feature vector $V(\frac{\Delta x}{w}, \frac{\Delta l}{h}, \frac{L_{max}}{h}, \frac{y_{low}}{h}, \frac{y_{high}}{h}, \frac{w_{p1}}{w}, \frac{w_{p2}}{w})$ according to formula

$$\begin{cases} \Delta x = |P1_x - P2_x| \\ \Delta l = |L_{p1} - L_{p2}| \\ L_{max} = \max(L_{p1}, L_{p2}) \\ y_{low} = \min(P1_yl, P2_yl) \\ y_{high} = \max(P1_yh, P2_yh) \end{cases} \quad (7)$$

where h is the height of f_b , Pi_yl and Pi_yh are low vertical coordinate and high vertical coordinate of candidate goal post Pi .

end

3. Highlight extraction and classification

3.1. Semantic boundary determination of highlight

The evolution of excitement level of soccer game could be directly revealed from the reactions of audience, commentator, video director. Hanjalic [12] first proposed to detect sports video highlights from the perspective of human affection. Different with Hanjalic's work, four affective factors namely STE, shot intensity (SI), shot change density (SCD) and replay factor (RF) are fused to generate highlight time curve (HTC). SCD is constructed as $SCD(i) = e^{(1-n(i))/r}$, here, $n(i)$ is the frame number of the shot which includes frame i . The parameter r is a constant empirically set to 300, by which the $SCD(i)$ values are scaled between 0 and 1. RF is constructed as $RF(k_i) = \frac{L(k)}{L_{const}} e^{-\frac{(i-k_0)^2}{L(k)^2 \cdot \alpha}}$, here, $L(k)$ is the frame number of replay k , $RF(k_i)$ denotes the replay factor value of the i -th frame in replay k , $i \in [-L(k), L(k)]$. L_{const} is empirically set to 2000, which scales $L(k)$ between 0 and 1. The shape parameter α is set to 5. After smoothed by Kaiser Window, these four affective factors are then linearly fused to generate HTC.

In soccer videos, the highlights always start from a FVS and end with a CUS. Usually, the FVS provides a global view of the whole field, while the CUS always focus on an action or facial expression of the player, coach or referee. In general, there is a replay following the CUS to recur the highlight. Such video editorial grammar is called event temporal transition pattern (ETTP), which is employed to determine the semantic boundary of highlight. The crests of HTC are first detected and filtered, then for each crest, the first FVS before the crest is determined as the start semantic boundary of the highlight, and the first CUS follows the start FVS is determined as the end semantic

boundary of the highlight. As a reasonable optional choice, the end frame of the replay which follows the start semantic boundary could also be thought as the end semantic boundary of the highlight. The former boundary is called short boundary, the latter boundary is called long boundary.

3.2. SVM based highlight classification

We first extract highlight feature vectors (HFVs) from the highlights with long boundaries. Based on the extensively investigations of soccer video highlights, 8 observations are considered as the distinguishable features, which are described as follows.

Highlight duration (D_h). The duration of highlights with long boundaries (in seconds). The highlight segment that contains goal event is usually longer.

Excitement level of highlight (EL_h). The average excitement levels of different types of highlights are usually different. Generally speaking, the goal event has the highest excitement level, next is the shoot event, foul event has the lowest excitement level.

Whistle in highlight (W_h). Whistle plays the important role in foul or set-piece kick event in soccer game.

Goalmouth in highlight (G_h). A binary (Yes/No) feature component. The goalmouth often appears in the shoot or goal events.

Replay duration (D_r). The duration of replay contained in highlights with long boundaries (in seconds). There is usually a 30 seconds or longer replay to describe the interesting events in soccer video.

Number of shots before replay (NS_{br}). If the goal event happens, the video director would use rich shot types such as MVS, CUS, and OFS to express the celebrations of players, coaches, and audiences. However, the foul event, which is usually an emergency, has smaller NS_{br} .

Number of shots in replay (NS_{ir}). In order to recur the details of foul events, the video director would frequently switch the shot, and hence, the number of shots in replay of foul event is often no less than 3.

FVS in replay (F_{ir}). A binary (Yes/No) feature component. FVS is seldom appeared in the replay of foul event.

After the set of feature components are calculated, the highlight could be represent as a feature vector, i.e., $HFV = \{D_h, EL_h, W_h, G_h, D_r, NS_{br}, NS_{ir}, F_{ir}\}$, which is used as the input of highlight classification model.

As the unique advantages in the pattern classification problem with characteristics of small samples, non-linear, and high-dimensions, the "one-against-all" SVM is trained to classify highlights into three types, namely goal, shoot and foul.

4. Experiments and evaluations

In this section, we evaluate the performance of whistle and goalmouth detection algorithm, and the performance of highlight classification using the recall R , and precision P , which are defined as follows:

$$R = \frac{T_p}{T_p + F_n} \times 100\% \quad (8)$$

$$P = \frac{T_p}{T_p + F_p} \times 100\% \quad (9)$$

where T_p , F_p , and F_n represent the true positive samples, false positive samples, and false negative samples, respectively.

4.1. Whistle detection

The proposed HT based whistle detection algorithm is compared with continuous hidden Markov model (CHMM) based method. The input features include 1-d ZCR, 1-d STE, 12-d MFCC coefficients, and 12-d MFCC difference coefficients. We compared the two methods on a 49 minutes soccer video and table 1 shows the results. We can find that the proposed HT based whistle detection algorithm achieves more effective performance.

Table 1. Comparison of whistle detection.

Method	Correct	False	Missed	<i>P</i>	<i>R</i>
HT	54	1	9	98.2%	85.7%
CHMM	49	8	14	86.0%	77.8%

4.2. Goalmouth detection

In the training phase of SVM, we manually collect 400 goalmouth images with various camera perspectives and distances from soccer videos of different leagues, as positive set, and 600 non-goalmouth images, for example, midfield, player with white clothes, the billboards on the sidelines and so on, as negative set. The kernel type of SVM is RBF.

In the testing phase of SVM, two 50-minutes soccer video from the South Africa world cup 2010 and UEFA Champions League are used to validate the proposed goalmouth detection algorithm. We detect the goalmouth every 10 frames, the results are shown in Table 2. In some cases, the players shield the goalmouth post, or only parts of the goalmouth are recorded by camera, so, a few of goalmouth images are missed. On the other hand, after the series of image processing steps, the joint effects of the white billboards, players with white clothes and the white field lines result in similar characteristics of goalmouth, which affect the precision rate.

Table 2. Performance of goalmouth detection algorithm.

Video	Correct	False	Missed	<i>P</i>	<i>R</i>
1	121	8	2	93.8%	98.4%
2	121	15	1	89.0%	99.2%
Total	242	23	3	91.3%	98.8%

4.3. Highlight classification

To evaluate the propose highlight classification approach, both the "one-against-all" SVM classifier and the three layer back propagation neural network (BPNN) classifier are modeled to classify the highlights into goal, shoot and foul.

The training and testing sets are collected from soccer videos of different leagues and manually annotated. The number of training and testing samples are listed in Table 3. As goal is the rare event in soccer game, the number of goal samples is less than other events.

Table 3. Training and testing sets for SVM and BPNN.

Highlights	Training samples	Testing samples
Goal	74	26
Shoot	688	208
Foul	396	95

The kernel function of SVM classifier is RBF. The activation function of BPNN is the sigmoid function, and the number of hidden layer neurons of BPNN is empirically set to 17. Table 4 and Table 5 show the confusion matrix of SVM based highlight classification approach, and BPNN based highlight classification approach, respectively. The shoot events are easy falsely labeled as goal events, which cause the lower precision and recall rate of goal events. Although the overall performance of SVM based highlight classification is a little superior to BPNN based highlight classification, the performance difference between the two classifiers is very small, which also demonstrates effectiveness of the proposed highlight classification framework.

Table 4. Confusion matrix of SVM based highlight classification.

	Goal	Shoot	Foul	<i>R</i>
Goal	23	3	0	88.5%
Shoot	4	199	5	95.7%
Foul	1	8	86	90.5%
<i>P</i>	82.1%	94.8%	94.5%	

Table 5. Confusion matrix of BPNN based highlight classification.

	Goal	Shoot	Foul	<i>R</i>
Goal	23	3	0	88.5%
Shoot	4	197	7	94.7%
Foul	1	10	84	88.4%
<i>P</i>	82.1%	93.8%	92.3%	

5. Conclusions

In this work, we combine the audio/visual features to assist semantic analysis of soccer video. The highlights with semantic boundaries are extracted based on HTC, which is generated by effective fusion of affection arousal factors. The HFVs are constructed to classify the highlights into goal, shoot and foul. The experiments show that, the proposed highlight classification approach is feasible and accurate. The value of excitement level computed by HTC could be utilized to rank the highlights. The fruits of this work could be further applied to event retrieval and user-oriented summarization. However, there are many types of highlights in soccer video, how to learn richer semantics is still a challenging problem. Next, we plan to exploit more effective representations to boost the event detection performance and applied to other sports videos.

Acknowledgement

This work is financially supported by Jiaxing Science and Technology Plan Project of China under Grant No. 2017AY13003 and Zhejiang Provincial Natural Science Foundation of China under Grant No. LY18F020021.

References

1. P. Oskouie, S. Alipour et al., Multimodal feature extraction and fusion for semantic mining of soccer video: a survey. *Artificial Intelligence Review*, 1-38, 2012.
2. X. Qian, H. Wang, et al., HMM based soccer video event detection using enhanced mid-level semantic. *Multimedia Tools and Applications*, 60:233-255, 2012.
3. D. W. Tjondronegoro and Y. P. P. Chen. Knowledge-discounted event detection in sports video. *IEEE Transactions on Systems Man and Cybernetics Part a-Systems and Humans*, 40(5):1009-1024, 2010
4. C. S. Xu, J. J. Wang, et al., A novel framework for semantic annotation and personalized retrieval of sports video. *IEEE Transactions on Multimedia*, 10(3):421-436, 2008.
5. A. A. Halin, M. Rajeswari and M. E. Abbasnejad, Soccer event detection via collaborative multimodal feature analysis and candidate ranking. *The International Arab Journal of Information Technology*, 10(5):1-9, 2013.
6. M. Xu, N. C. Maddage, et al., Creating audio keywords for event detection in soccer video. In *proceedings of International Conference on Multimedia Expo (ICME)*, 281-284, 2003.
7. M. Xu, C. S. Xu, et al., Audio keywords generation for sports video analysis. *ACM Transactions on Multimedia Computing, Communications and Applications*, 4(2):11-23, 2008.
8. Y. Zhang, J. Wang, and Y. Dai, Soccer video shot segmentation based on self-adapting dual threshold and dominant color percentage. *Journal of Nanjing University of Science and Technology (Natural Science)*, 33(4):432-437, 2009.
9. J. Q. Yu, N. Wang. Shot classification for soccer video based on sub-window region. *Journal of Image and Graphics*, 13(7):1347-1352, 2008.
10. Q. Huang, J. Hu, et al., A reliable logo and replay detector for sports video. In *Proceedings of IEEE International Conference on Multimedia and Expo (ICME)*, 1695-1698, 2007.

11. J. Q. Yu, H. H. He and Y. F. He, Highlights extraction for soccer video based on affection arousal. *Journal of Computer Research and Development*, 47(10):1823-1831, 2010.
12. Hanjalic. Adaptive extraction of highlights from a sport video based on excitement modeling. *IEEE Transactions on Multimedia*, 7(6):1114-1122, 2005.